Lesson 36: Repeated Games and Folk Theorems

Curriculum: A Rigorous Learning Path to Multi-Agent Deep RL

Contents

1 Introduction **2**

2 Finitely Repeated Games **3**
- 2.1 Model . . . 3
- 2.2 Subgame Perfect Equilibrium . . . 3
- 2.3 The Unraveling Argument . . . 3
- 2.4 Multiple Stage-Game Equilibria . . . 4

3 Infinitely Repeated Games **5**
- 3.1 Model . . . 5
- 3.2 Feasible and Individually Rational Payoffs . . . 5
- 3.3 Trigger Strategies . . . 6

4 Folk Theorems **6**
- 4.1 Nash Folk Theorem . . . 7
- 4.2 Subgame Perfect Folk Theorem . . . 8

1 Introduction

In Lesson 11 we studied one-shot (normal-form) games: simultaneous moves, Nash equilibrium, and the minimax theorem. However, most real strategic interactions—and virtually all multi-agent reinforcement learning (MARL) scenarios—involve *repeated interaction*. Agents meet the same opponents across many episodes, observe each other's past behavior, and adapt their strategies over time. Three fundamental phenomena emerge:

(i) **Cooperation through repetition.** Even when one-shot incentives favor defection (as in the Prisoner's Dilemma), the threat of future punishment can sustain cooperative outcomes. The *folk theorems* characterize exactly which payoff vectors are achievable.

(ii) **Incomplete information.** In practice, agents rarely know each other's payoff functions. A robot negotiating with a human does not know the human's utility; competing firms do not know rivals' costs. *Bayesian games* model this uncertainty, and *perfect Bayesian equilibrium* handles sequential decision-making under such uncertainty.

(iii) **Structured interactions.** Many multi-agent problems possess special structure—such as *potential games*, where a single global function captures all players' incentives. In these settings, simple learning rules (best-response dynamics, log-linear learning) are guaranteed to converge.

This document covers all three themes with complete proofs. The roadmap is as follows.

- **Section 2**: Finitely repeated games and the unraveling argument.
- **Section 3**: Infinitely repeated games, trigger strategies, and feasible/individually rational payoffs.
- **Section 4**: The Nash folk theorem and the subgame perfect folk theorem.
- **Section ??**: Fictitious play, no-regret learning, and convergence in zero-sum games.
- **Section ??**: Bayesian games and Bayesian Nash equilibrium.
- **Section ??**: Perfect Bayesian equilibrium and signaling games.
- **Section ??**: Potential games, congestion games, and convergence of best-response dynamics.
- **Section ??**: Connections to multi-agent RL.
- **Section ??**: Exercises.
- **Section ??**: Summary of key results.

Why This Matters for MARL

In multi-agent RL, agents interact repeatedly and update their policies based on observed outcomes. The folk theorems explain why independent Q-learners can converge to cooperative (or adversarial) outcomes that do not arise in single-episode analysis. Bayesian games formalize the partial observability inherent in opponent modeling. Potential games identify problem classes where independent learning is guaranteed to converge—a rare and valuable property in general-sum MARL.

2 Finitely Repeated Games

2.1 Model

We begin with a *stage game* $G = (N, (A_i)_{i\in N}, (g_i)_{i\in N})$ where $N = [n]$ is the player set, A_i is the finite action set of player i, and $g_i : A \to \mathbb{R}$ is the stage-game payoff function, with $A = \prod_{i\in N} A_i$.

Definition 2.1 (Finitely Repeated Game)**.** The *T-fold repetition* of G, denoted G^T, is defined as follows:

(a) In each period $t \in \{1, 2, \ldots, T\}$, all players simultaneously choose actions $a_i^t \in A_i$.

(b) At the end of period t, all players observe the realized action profile $a^t = (a_1^t, \ldots, a_n^t)$.

(c) The *history* at period t is $h^t = (a^1, a^2, \ldots, a^{t-1}) \in A^{t-1}$. The initial history is $h^1 = \emptyset$.

(d) A *(pure) strategy* for player i is a sequence of functions $s_i = (s_i^1, s_i^2, \ldots, s_i^T)$ where $s_i^t : A^{t-1} \to A_i$ maps histories to actions.

(e) The payoff to player i under strategy profile $s = (s_1, \ldots, s_n)$ is

$$u_i(s) = \frac{1}{T}\sum_{t=1}^{T} g_i\big(a^t(s, h^t)\big),$$

where $a^t(s, h^t) = \big(s_1^t(h^t), \ldots, s_n^t(h^t)\big)$ and histories are generated by the strategy profile.

Remark 2.1. We use average payoffs $\frac{1}{T}\sum_t g_i(a^t)$ rather than total payoffs to keep the repeated-game payoffs on the same scale as stage-game payoffs, facilitating comparison with the infinite-horizon discounted case.

2.2 Subgame Perfect Equilibrium

The repeated game G^T is an extensive-form game of perfect information (after each period, all actions are observed). Thus the natural solution concept is *subgame perfect equilibrium* (SPE): a strategy profile that constitutes a Nash equilibrium in every subgame.

Definition 2.2 (Subgame Perfect Equilibrium)**.** A strategy profile s^* in G^T is a *subgame perfect equilibrium* (SPE) if, for every period $t \in \{1, \ldots, T\}$ and every history $h^t \in A^{t-1}$, the continuation strategy profile $(s_i^{*,\tau}(h^t, \cdot))_{\tau\geq t}$ is a Nash equilibrium of the subgame starting from h^t.

2.3 The Unraveling Argument

The fundamental result for finitely repeated games with a unique stage-game equilibrium is that repetition adds nothing: the unique SPE plays the stage-game NE in every period.

Theorem 2.1 (Unique NE Implies Unique SPE)**.** *If the stage game G has a unique Nash equilibrium $a^* = (a_1^*, \ldots, a_n^*)$, then the T-fold repeated game G^T has a unique subgame perfect equilibrium in which $a^t = a^*$ for all $t \in \{1, \ldots, T\}$.*

Proof. We proceed by backward induction on t.

Base case ($t = T$): In the final period, regardless of history h^T, the subgame is identical to the one-shot stage game G. Since a^* is the unique NE of G, every SPE must play a^* in period T.

Inductive step: Suppose that in every SPE, $a^\tau = a^*$ for all $\tau \in \{t + 1, \ldots, T\}$, regardless of the history at period $t + 1$. Consider the subgame beginning at an arbitrary history h^t.

The continuation payoffs from period $t+1$ onward are fixed at $\frac{1}{T}\sum_{\tau=t+1}^{T} g_i(a^*) = \frac{T-t}{T} g_i(a^*)$, independent of the action chosen at period t (since all future play is a^* regardless of history). Therefore, player i's effective payoff at period t is

$$\frac{1}{T} g_i(a^t) + \frac{T-t}{T} g_i(a^*),$$

and maximizing this over a_i^t is equivalent to maximizing $g_i(a_i^t, a_{-i}^t)$ in the stage game. Since a^* is the unique NE, $a^t = a^*$ in every SPE.

By induction, $a^t = a^*$ for all t, proving that the unique SPE is to play a^* in every period. □

The Unraveling Trap

Theorem 2.1 shows that finite repetition of the Prisoner's Dilemma (which has a unique NE at mutual defection) cannot sustain cooperation under subgame perfection. This is counterintuitive and motivates the study of infinitely repeated games, where the "last period" argument cannot be applied.

2.4 Multiple Stage-Game Equilibria

When the stage game has multiple Nash equilibria, repetition *can* create new equilibrium outcomes.

Proposition 2.2 (New Equilibria Through Repetition)**.** *If the stage game G has at least two Nash equilibria yielding different payoffs for some player, then the two-fold repeated game G^2 may have SPE outcomes in which first-period play is not a stage-game NE.*

Proof. We give a constructive proof. Let G have two Nash equilibria a^* and $\hat{a}$ with $g_i(a^*) > g_i(\hat{a})$ for all i (i.e., a^* Pareto-dominates $\hat{a}$). Suppose there exists an action profile $\tilde{a}$ (not necessarily a NE) with $g_i(\tilde{a}) > g_i(a^*)$ for all i, and suppose the gain from deviating unilaterally in the stage game satisfies

$$\max_{a_i' \in A_i} g_i(a_i', \tilde{a}_{-i}) - g_i(\tilde{a}) < g_i(a^*) - g_i(\hat{a}).$$

Consider the following strategy for each player i in G^2:

- Period 1: Play $\tilde{a}_i$.
- Period 2: If $a^1 = \tilde{a}$, play a_i^*. Otherwise, play $\hat{a}_i$.

In period 2, players play a stage-game NE (a^* or $\hat{a}$) depending on period-1 history, so the strategy is sequentially rational in period 2.

In period 1, if player i deviates to a_i' while others play $\tilde{a}_{-i}$, the payoff change is:

$$\begin{aligned}
&\frac{1}{2}\big[g_i(a_i', \tilde{a}_{-i}) + g_i(\hat{a})\big] - \frac{1}{2}\big[g_i(\tilde{a}) + g_i(a^*)\big] \\
&= \frac{1}{2}\big[g_i(a_i', \tilde{a}_{-i}) - g_i(\tilde{a})\big] - \frac{1}{2}\big[g_i(a^*) - g_i(\hat{a})\big] < 0,
\end{aligned}$$

by the assumed inequality. Thus no player wants to deviate, and the profile is an SPE in which first-period play $\tilde{a}$ is *not* a stage-game NE. □

Example 2.1 (Coordination Game)**.** Consider the two-player coordination game with payoffs:

	L	R
T	$3,3$	$0,0$
B	$0,0$	$1,1$

This has two pure NE: (T, L) with payoffs $(3,3)$ and (B, R) with payoffs $(1,1)$. In the two-fold repetition, the "punishment" equilibrium (B, R) can incentivize cooperation on outcomes that are not stage-game NE, provided the incentive condition is satisfied.

3 Infinitely Repeated Games

3.1 Model

To circumvent the unraveling argument, we consider infinite repetition with discounting.

Definition 3.1 (Infinitely Repeated Game)**.** Given a stage game G and a discount factor $\delta \in (0,1)$, the *infinitely repeated game* $G^\infty(\delta)$ is defined by:

(a) In each period $t \in \{1, 2, 3, \ldots\}$, players simultaneously choose actions $a^t \in A$.

(b) Histories, strategies are defined as in the finite case, but with t ranging over $\mathbb{N}$.

(c) Player i's payoff under strategy profile s is the *normalized discounted sum*:

$$u_i(s) = (1-\delta)\sum_{t=1}^{\infty} \delta^{t-1} g_i(a^t).$$

The normalization factor $(1-\delta)$ ensures that payoffs lie in the same range as stage-game payoffs: if $a^t = a$ for all t, then $u_i = g_i(a)$.

3.2 Feasible and Individually Rational Payoffs

Definition 3.2 (Feasible Payoff Set)**.** The *feasible payoff set* of the stage game G is

$$V = \operatorname{conv}\{g(a) : a \in A\} = \operatorname{conv}\{(g_1(a), \ldots, g_n(a)) : a \in A\} \subset \mathbb{R}^n,$$

the convex hull of all achievable stage-game payoff vectors.

Remark 3.1. In the infinitely repeated game, any payoff vector $v \in V$ is achievable by cycling through action profiles with appropriate frequencies. For instance, if $v = \lambda g(a) + (1-\lambda) g(a')$ for some $\lambda \in [0,1]$, then playing a for a fraction λ of periods and a' for the remaining fraction achieves payoff v in the limit as $\delta \to 1$.

Definition 3.3 (Minimax Value and Individually Rational Payoffs)**.** The *minimax value* (or *minmax value*) of player i in pure strategies is

$$\underline{v}_i = \min_{a_{-i} \in A_{-i}} \max_{a_i \in A_i} g_i(a_i, a_{-i}).$$

The *mixed-strategy minimax value* is

$$\underline{v}_i^{\text{mix}} = \min_{\alpha_{-i} \in \Delta(A_{-i})} \max_{a_i \in A_i} \mathbb{E}_{\alpha_{-i}}[g_i(a_i, a_{-i})],$$

where $\Delta(A_{-i}) = \prod_{j \neq i} \Delta(A_j)$ is the set of independent mixed strategy profiles for i's opponents. A payoff vector $v \in \mathbb{R}^n$ is *individually rational* if $v_i \geq \underline{v}_i^{\text{mix}}$ for all i.

Proposition 3.1. *In any Nash equilibrium of $G^\infty(\delta)$, each player i receives payoff $u_i \geq \underline{v}_i^{mix}$.*

Proof. In any NE, player i can deviate to the strategy of playing $\operatorname{argmax}_{a_i} \mathbb{E}_{\alpha_{-i}^t}[g_i(a_i, a_{-i}^t)]$ in every period t (where α_{-i}^t is the marginal distribution of opponents' play at time t induced by the equilibrium). This guarantees a stage payoff of at least $\underline{v}_i^{\text{mix}}$ in every period, hence a total payoff of at least $\underline{v}_i^{\text{mix}}$. Since the NE strategy must yield payoff at least as high as any deviation, we get $u_i \geq \underline{v}_i^{\text{mix}}$. □

3.3 Trigger Strategies

Definition 3.4 (Grim Trigger Strategy)**.** Given a target action profile $a^* \in A$, the *grim trigger strategy* for player i is:

$$s_i^t(h^t) = \begin{cases} a_i^* & \text{if } a^\tau = a^* \text{ for all } \tau < t, \\ m_i & \text{otherwise,} \end{cases}$$

where $m_i \in \operatorname{argmin}_{\alpha_i} \max_{a_j} g_j(a_j, \alpha_{-j})$ is player i's component of a minimax punishment profile against any deviator.

Example 3.1 (Cooperation in the Repeated Prisoner's Dilemma)**.** Consider the Prisoner's Dilemma with payoffs:

	C	D
C	$3,3$	$0,4$
D	$4,0$	$1,1$

The unique stage-game NE is (D, D) with payoffs $(1, 1)$. The minimax value for each player is $\underline{v}_i = 1$. Consider the grim trigger strategy profile that plays (C, C) and punishes with (D, D) forever.

Cooperation payoff: If both cooperate forever, each gets $u_i = 3$.

Deviation payoff: If player i defects in period t (while the opponent cooperates), she gets 4 in that period and 1 thereafter:

$$u_i^{\text{dev}} = (1-\delta)\big[\delta^{t-1} \cdot 4 + \sum_{\tau=t+1}^{\infty} \delta^{\tau-1} \cdot 1\big] + (1-\delta) \sum_{\tau=1}^{t-1} \delta^{\tau-1} \cdot 3.$$

The best deviation is at $t = 1$:

$$u_i^{\text{dev}} = (1-\delta) \cdot 4 + \delta \cdot 1 = 4 - 3\delta.$$

Cooperation is sustained iff $3 \geq 4 - 3\delta$, i.e., $\delta \geq \frac{1}{3}$.

Discount Factors and Episode Length in RL

The discount factor δ in repeated games is analogous to the discount factor γ in MDPs. In MARL, if γ is too small (agents are too myopic), cooperative equilibria may not be sustainable. This directly parallels the threshold $\delta \geq 1/3$ in the repeated Prisoner's Dilemma: myopic agents (small γ) cannot learn to cooperate.

Definition 3.5 (Tit-for-Tat)**.** In a two-player game, the *tit-for-tat* strategy for player i is:

$$s_i^1 = C, \qquad s_i^t(h^t) = a_j^{t-1} \quad \text{for } t \geq 2,$$

where $j \neq i$. That is, start by cooperating, then mimic the opponent's previous action.

Remark 3.2. Tit-for-tat is a "forgiving" strategy: unlike grim trigger, punishment is temporary. Axelrod's famous tournaments showed that tit-for-tat performs well in evolutionary settings, which connects to the multi-agent RL idea that simple, interpretable strategies can be surprisingly effective.

4 Folk Theorems

The folk theorems characterize the equilibrium payoff set of infinitely repeated games. Informally, "almost anything" can be sustained as an equilibrium if players are patient enough.

4.1 Nash Folk Theorem

Theorem 4.1 (Nash Folk Theorem). *Let G be a finite normal-form game and $V = \text{conv}\{g(a) : a \in A\}$ the feasible payoff set. For any payoff vector $v \in V$ satisfying $v_i > \underline{v}_i^{mix}$ for all i (strictly individually rational), there exists $\underline{\delta} \in (0,1)$ such that for all $\delta \in (\underline{\delta}, 1)$, there is a Nash equilibrium of $G^\infty(\delta)$ with payoff vector v.*

Proof. We construct a Nash equilibrium yielding payoff v.

Step 1: Achieving v through action sequences. Since $v \in V = \text{conv}\{g(a) : a \in A\}$, there exist action profiles $a(1), a(2), \ldots, a(K) \in A$ and weights $\lambda_1, \ldots, \lambda_K \geq 0$ with $\sum_k \lambda_k = 1$ such that $v = \sum_k \lambda_k g(a(k))$. We can approximate v by cycling through these action profiles. Specifically, for any $\varepsilon > 0$ and δ close enough to 1, we can construct a deterministic sequence of action profiles $(a^t)_{t=1}^\infty$ such that the discounted average payoff $(1-\delta)\sum_{t=1}^\infty \delta^{t-1} g(a^t)$ is within ε of v.

More precisely, define a cycle of length L where action profile $a(k)$ is played for $\lceil \lambda_k L \rceil$ periods (adjusting the last to make the total exactly L). As $\delta \to 1$, the discounted average of any periodic sequence converges to the time-average, which converges to v as $L \to \infty$. Fix $\varepsilon > 0$ small enough that $v_i - \varepsilon > \underline{v}_i^{\text{mix}}$ for all i, and choose L and $\underline{\delta}_1$ accordingly.

Step 2: Punishment strategies. For each player i, let $\alpha_{-i}^{m,i} \in \Delta(A_{-i})$ be a mixed minimax punishment profile: $\max_{a_i} \mathbb{E}_{\alpha_{-i}^{m,i}}[g_i(a_i, a_{-i})] = \underline{v}_i^{\text{mix}}$. In the punishment phase, players $j \neq i$ play according to $\alpha_{-i}^{m,i}$ independently.

Step 3: Strategy construction. Define the following strategy for each player j:

- *Cooperation phase:* Play the sequence $(a^t)_{t=1}^\infty$ from Step 1.
- *Punishment phase:* If some player i is the first to deviate from the prescribed sequence in some period $\hat{t}$, then for all $t > \hat{t}$, all players $j \neq i$ play according to $\alpha_{-i}^{m,i}$ (punishing i forever).

Step 4: Verifying the Nash equilibrium condition. Consider whether player i wants to deviate. If player i follows the prescribed strategy, she receives payoff within ε of v_i (for $\delta > \underline{\delta}_1$).

If player i deviates at any period $\hat{t}$, she receives at most $\max_a g_i(a) \cdot (1-\delta)\delta^{\hat{t}-1}$ in that period, plus the punishment continuation payoff. In the punishment phase, the opponents play the mixed minimax strategy, so player i's best response yields at most $\underline{v}_i^{\text{mix}}$ per period. Thus the deviation payoff is at most:

$$u_i^{\text{dev}} \leq (1-\delta)\sum_{\tau=1}^{\hat{t}} \delta^{\tau-1} M + \delta^{\hat{t}} \underline{v}_i^{\text{mix}}$$

where $M = \max_a |g_i(a)|$.

As $\delta \to 1$, the first term (a finite number of periods) vanishes and $u_i^{\text{dev}} \to \underline{v}_i^{\text{mix}} < v_i - \varepsilon$. Therefore, for δ large enough (say $\delta > \underline{\delta}_2$), deviation is unprofitable.

Taking $\underline{\delta} = \max(\underline{\delta}_1, \underline{\delta}_2)$ completes the proof.

Note on Nash vs. SPE: This equilibrium is Nash but not necessarily subgame perfect: punishing players have no incentive to carry out the punishment (they are indifferent or may prefer not to punish). The subgame perfect folk theorem below addresses this. □

Nash Folk Theorem – Interpretation

Any feasible, strictly individually rational payoff vector is achievable as a Nash equilibrium payoff for sufficiently patient players. This means the set of NE payoffs of the repeated game "fills up" the feasible individually rational region as $\delta \to 1$.

4.2 Subgame Perfect Folk Theorem

The Nash folk theorem has a weakness: punishment threats may not be credible. The punishers might prefer to deviate from punishment. The subgame perfect folk theorem (Fudenberg and Maskin, 1986) resolves this by constructing self-enforcing punishment schemes.

Definition 4.1 (Full Dimensionality Condition)**.** The stage game G satisfies the *full dimensionality condition* if the feasible payoff set $V = \text{conv}\{g(a) : a \in A\}$ has dimension n (the number of players), i.e., V has nonempty interior in $\mathbb{R}^n$.

Theorem 4.2 (Subgame Perfect Folk Theorem (Fudenberg–Maskin))**.** *Let G be a finite n-player game satisfying the full dimensionality condition. For any feasible payoff vector $v \in V$ with $v_i > \underline{v}_i^{mix}$ for all i, there exists $\underline{\delta} \in (0,1)$ such that for all $\delta \in (\underline{\delta}, 1)$, there is a subgame perfect equilibrium of $G^\infty(\delta)$ with payoff vector v.*

Proof. The proof is constructive. We build a strategy profile with multiple phases: a cooperation phase and player-specific punishment phases, arranged so that each phase is self-enforcing.

Step 1: Setup. Fix $v \in V$ with $v_i > \underline{v}_i^{\text{mix}}$ for all i. Since V has full dimension, for each player i there exists a feasible payoff vector $w^i \in V$ with:

(i) $w_j^i > \underline{v}_j^{\text{mix}}$ for all $j \neq i$ (the punishers receive strictly above their minimax values during the punishment of i),

(ii) $w_i^i < v_i$ (player i's payoff during her own punishment phase is below her cooperative payoff; we can choose w_i^i close to $\underline{v}_i^{\text{mix}}$).

The full dimensionality condition ensures such w^i exist: since V has nonempty interior, we can find vectors that independently adjust each player's payoff.

Let $m_{-i}^i \in \Delta(A_{-i})$ be a mixed minimax strategy profile against player i.

Step 2: Strategy construction (three-phase strategy).

Phase 0 (Cooperation): Play the action sequence that yields payoff v (as in the Nash folk theorem proof).

Phase I_i (Punishment of player i): Lasts for L periods. During this phase, the opponents of i play the minimax strategy m_{-i}^i in each period, and player i plays any best response (which yields at most $\underline{v}_i^{\text{mix}}$ per period).

Phase II_i (Reward after punishment): After the L-period punishment phase, play switches to the action sequence yielding payoff w^i forever. This rewards the punishers (who received above their minimax during punishment) and keeps i's continuation payoff below v_i if L is large enough.

Transitions:

- If player j is the first to deviate in Phase 0, switch to Phase I_j.
- If player j deviates during Phase I_i (and $j \neq i$), restart Phase I_j (punish the deviating punisher).
- If player i deviates during her own punishment Phase I_i, restart Phase I_i.

Step 3: Choosing L. We need L large enough that deviation is unprofitable. Let $M = \max_{a,i} |g_i(a)|$. During Phase I_i, player i gets at most $\underline{v}_i^{\text{mix}}$ per period. After Phase I_i, player i gets w_i^i per period. Thus player i's continuation payoff from Phase I_i is:

$$(1-\delta)\sum_{t=1}^{L} \delta^{t-1}\underline{v}_i^{\text{mix}} + \delta^L w_i^i = (1-\delta^L)\underline{v}_i^{\text{mix}} + \delta^L w_i^i.$$

We need this to be less than v_i (so punishment deters deviation from Phase 0) and for the punishers' payoffs to make Phase I self-enforcing.

Step 4: Verifying sequential rationality.

Phase 0 – Player i considers deviating: Deviation yields at most M in one period, then Phase I_i begins. Compliance yields v_i forever. The deviation is unprofitable when:

$$v_i \geq (1-\delta)M + \delta\big[(1-\delta^L)\underline{v}_i^{\text{mix}} + \delta^L w_i^i\big].$$

As $\delta \to 1$, the right side approaches $(1-\delta)M + \delta[(1-\delta^L)\underline{v}_i^{\text{mix}} + \delta^L w_i^i]$. For δ close to 1 and L large, $(1-\delta)M \to 0$ and the continuation approaches a weighted average of $\underline{v}_i^{\text{mix}}$ and w_i^i, both strictly below v_i (since $w_i^i < v_i$ and $\underline{v}_i^{\text{mix}} < v_i$). So the inequality holds for δ large enough.

Phase I_i – Punisher $j \neq i$ considers deviating: If j deviates during Phase I_i, Phase I_j begins (punishing j). Compliance with Phase I_i yields: the minimax strategy payoff for $L-t$ remaining periods (at worst $-M$ per period but bounded), followed by $w_j^i > \underline{v}_j^{\text{mix}}$ forever. Deviation yields at most M for one period, then Phase I_j punishment.

For δ close to 1, the continuation payoff from compliance converges to $w_j^i > \underline{v}_j^{\text{mix}}$, while deviation leads eventually to w_j^j which can be chosen close to $\underline{v}_j^{\text{mix}}$. So compliance dominates for large δ.

Phase I_i – Player i considers deviating: Deviating restarts Phase I_i, gaining at most M in one period but adding L more punishment periods. Compliance ends the punishment sooner and leads to w_i^i forever. For large δ, the cost of extending punishment exceeds any one-period gain.

Since all deviations are unprofitable for $\delta > \underline{\delta}$ (where $\underline{\delta}$ depends on the game parameters $M, v, w^i, \underline{v}_i^{\text{mix}}, L$), the constructed strategy profile is a subgame perfect equilibrium. □

Remark 4.1. The full dimensionality condition is needed only for $n \geq 3$ players: it allows constructing punishment payoffs w^i that reward the punishers while punishing the deviator. For two-player games, Fudenberg and Maskin (1986) showed the folk theorem holds without this condition, since minmaxing the opponent is the only thing the punisher can do.

Subgame Perfect Folk Theorem – Interpretation

For patient enough players (large δ), the set of SPE payoffs of the infinitely repeated game is exactly the set of feasible, strictly individually rational payoff vectors (under the full dimensionality condition). This is a much stronger result than the Nash folk theorem because SPE requires credible (self-enforcing) punishments.

Folk Theorems and Multi-Agent RL

The folk theorems have profound implications for MARL:

(i) **Multiplicity of equilibria:** In repeated general-sum games, there are typically *many* equilibria. This means that independent learners may converge to vastly different outcomes depending on initial conditions, learning rates, and exploration schedules.

(ii) **Emergent cooperation:** The folk theorems show that cooperation can be sustained by punishment threats, explaining why self-play agents sometimes learn cooperative strategies in iterated social dilemmas.

(iii) **Reward shaping:** The discount factor δ (analogous to γ in RL) determines the set of achievable equilibria. Curriculum designers can influence emergent behavior by adjusting γ.